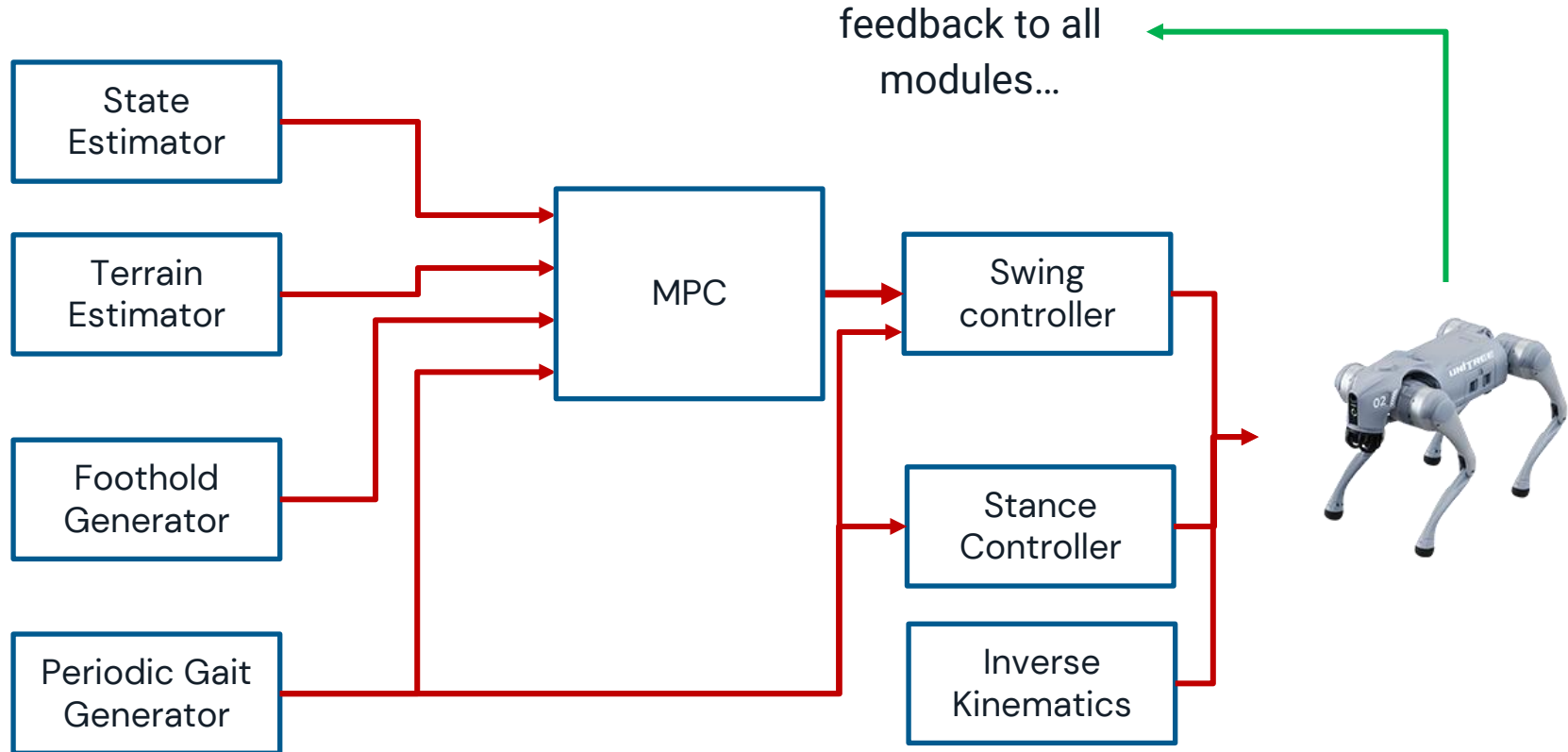


Lab Sessions I

Two lab sessions on **a complete model-based framework** for a legged quadruped:

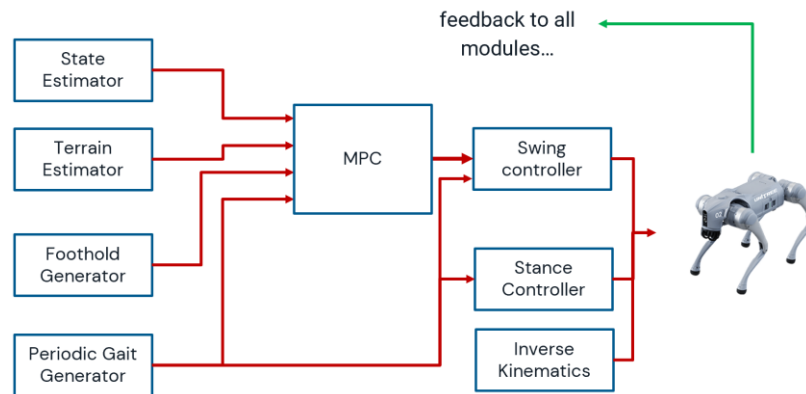
- Friday: Test in simulation
- Saturday: Test on a real robot



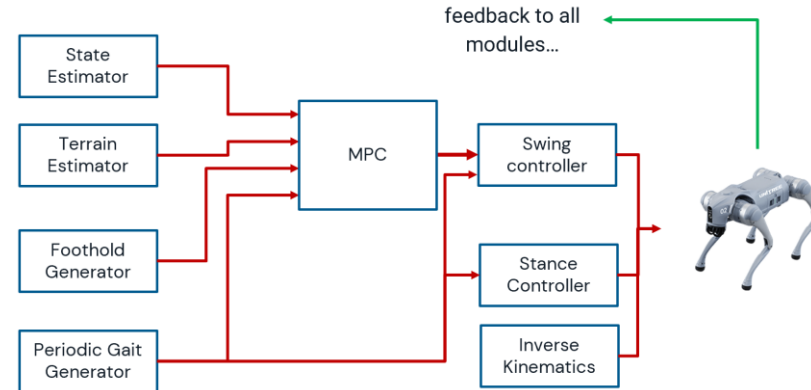


Kalman Filter which provides **state feedback**:

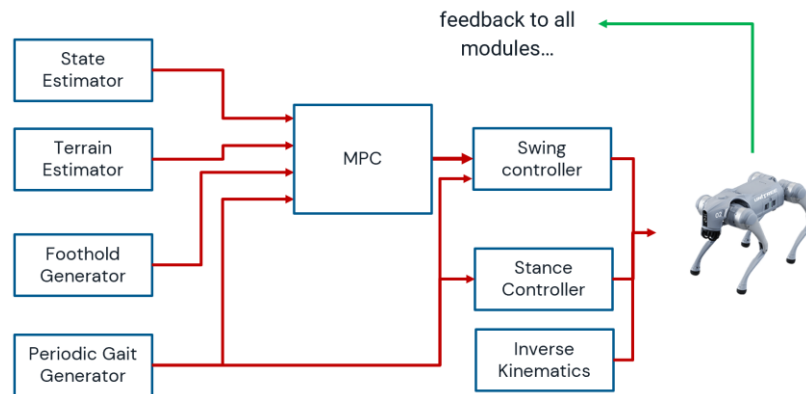
- base position
- base linear velocity
- base angular position
- base angular velocity



When the robot goes up the stairs, **how we should change its pitch and roll?**



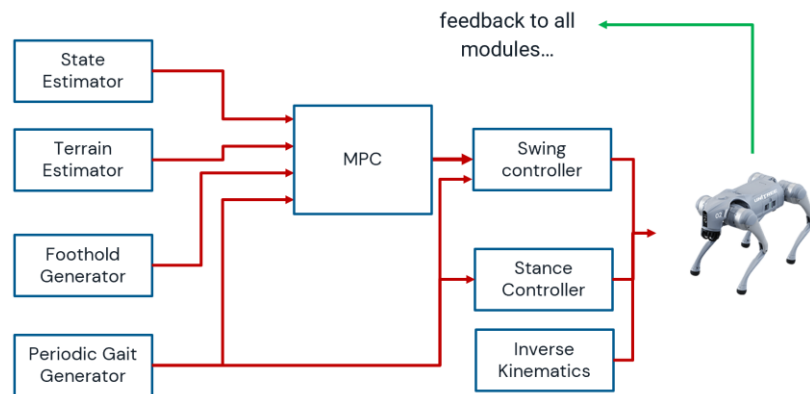
Where the robot should place **its feet**?



Periodic Gait Generator

Remember: model-based control usually precompute gait timing (e.g. when the foot should be on terrain – stance, and when in air - swing).

Which are the common gaits and how we generate them?



Trot



Crawl



Pace

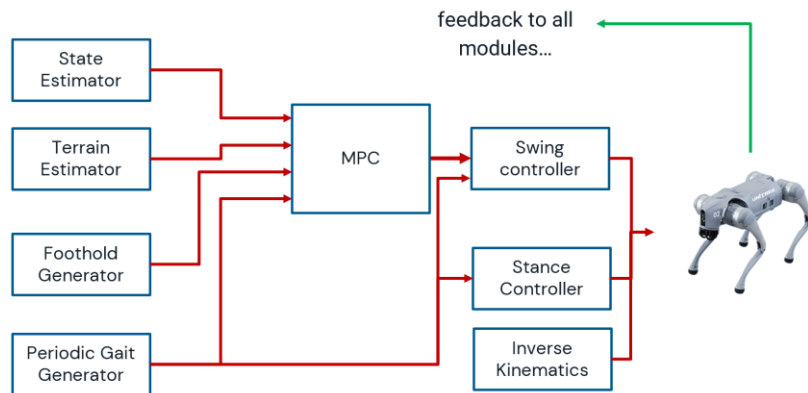
Periodic Gait Generator

Remember: model-based control usually precompute gait timing (e.g. when the foot should be on terrain – stance, and when in air - swing).

Which are the common gaits and how we generate them?



Trot

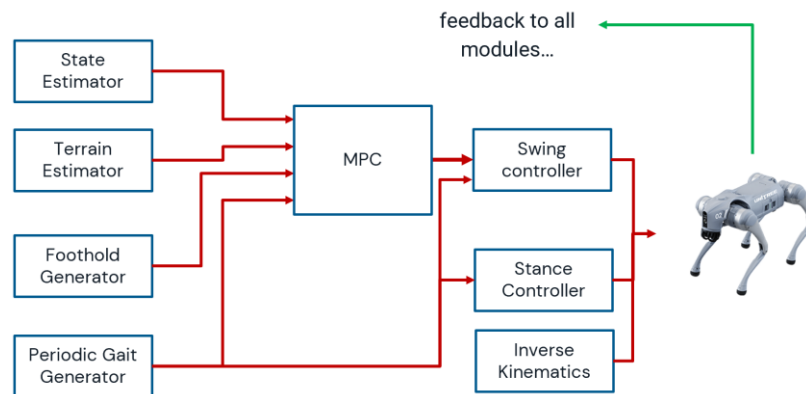


Front Left:	1	0	0	0	1	1	1	1	1
Front Right:	1	1	1	1	1	1	0	0	0
Rear Left:	1	1	1	1	1	1	0	0	0
Rear Right:	1	0	0	0	1	1	1	1	1

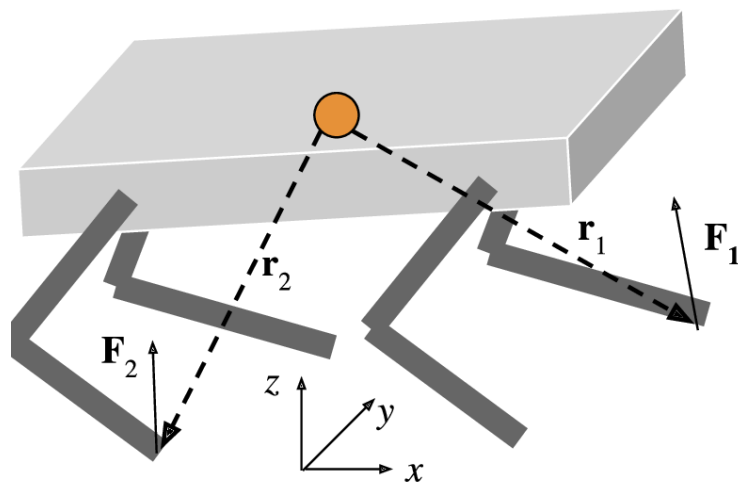
gait sequence

Based on:

- LIP
- **SRBD**
-
- Full Model



SRBD model:



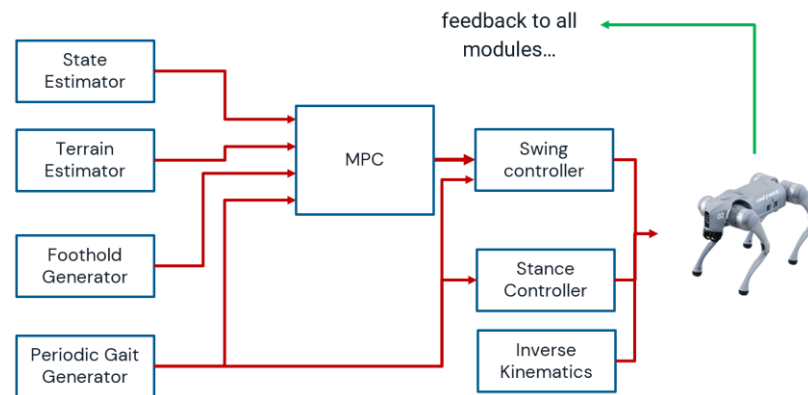
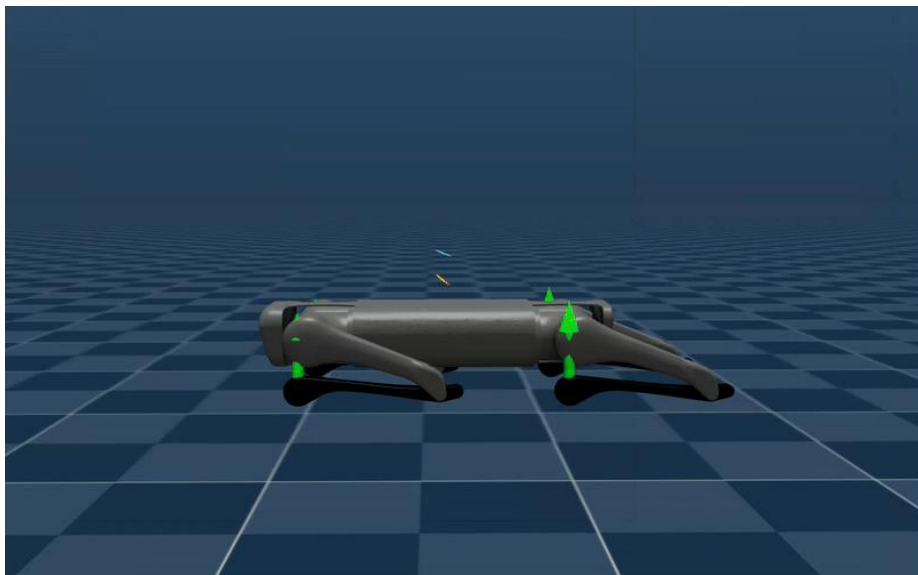
$$\mathbf{u} = [\mathbf{r}, \mathbf{F}] \quad \longrightarrow \quad \text{inputs}$$

$$m(\ddot{x}_{com} + g) = \sum_{i=1}^c F_i \delta_i \quad \longrightarrow \quad \text{linear part}$$

$$I_G \dot{\omega}_b \simeq \sum_{i=1}^c (F_i \delta_i \times r_i) \quad \longrightarrow \quad \text{angular part}$$

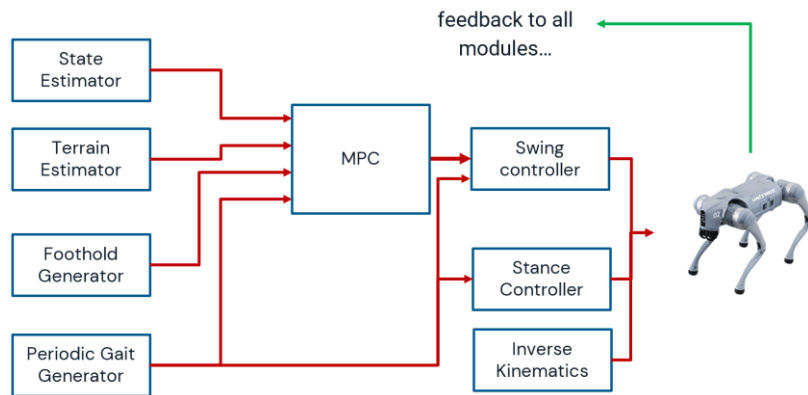
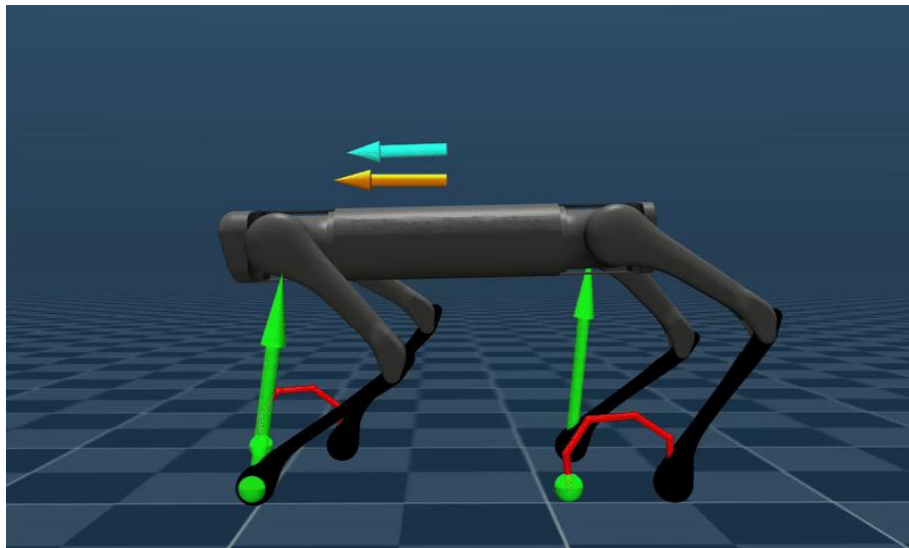
$$\delta \quad \longrightarrow \quad \text{Gait parameters}$$

When the MPC computes the GRF **for the feet in stance, how we map them into motor torques?**



Swing Controller

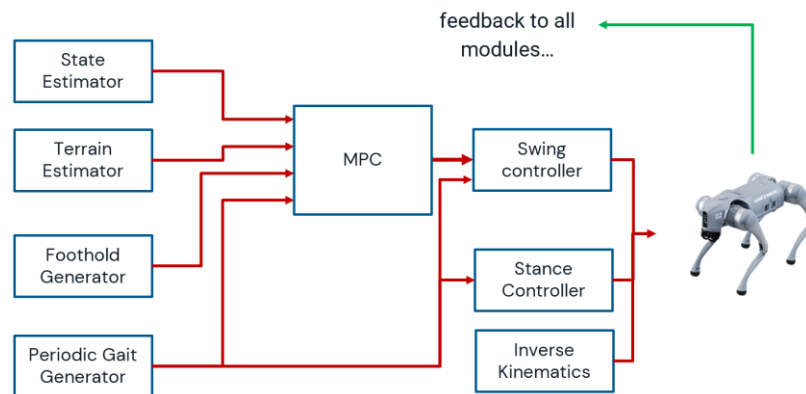
What happens instead to **the feet in air**?

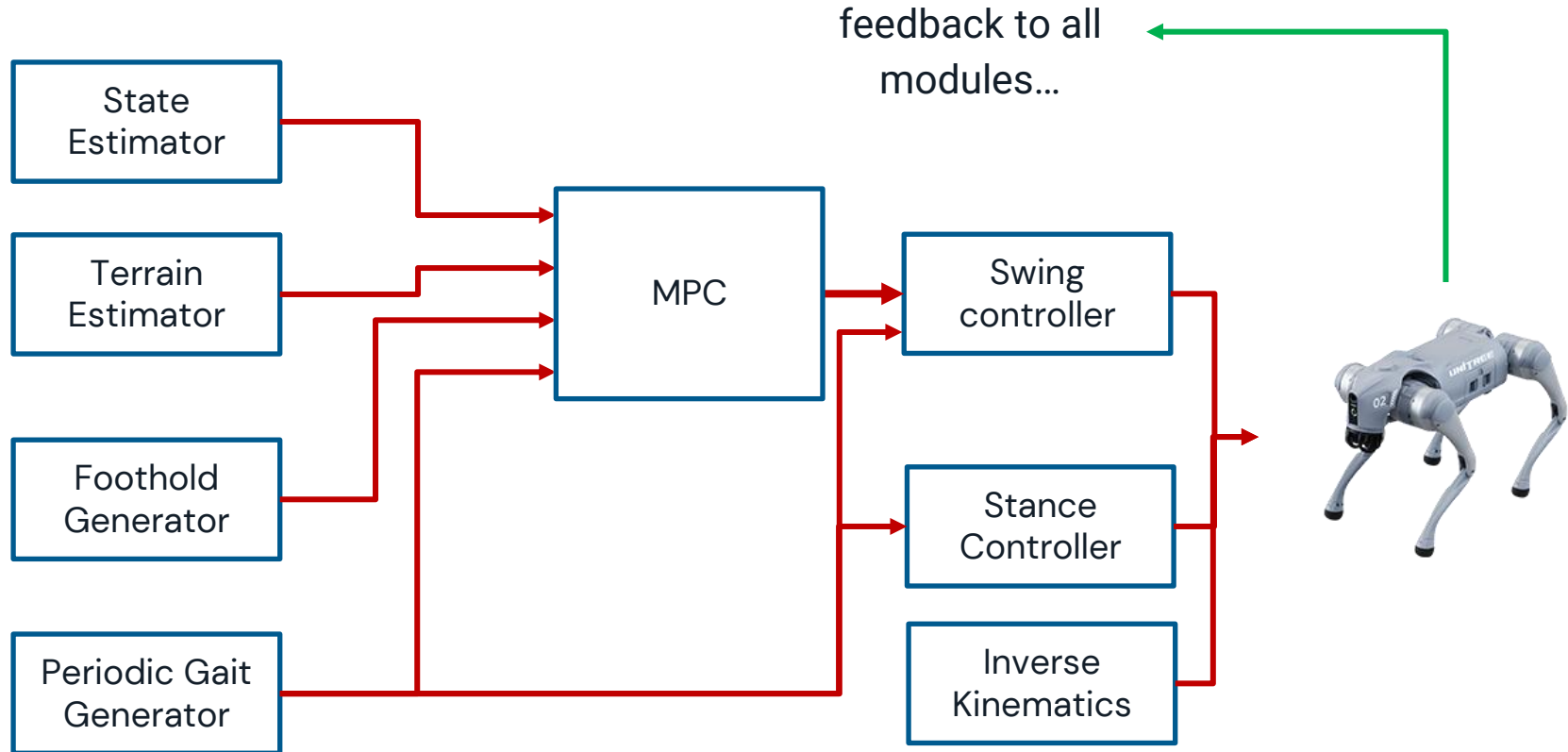


Real-robot have an internal closed loop, that is:

$$\tau_{ff} + K_p(q_{des} - q) + K_d(\dot{q}_{des} - \dot{q})$$

- τ_{ff} is called feed forward torque, came from swing and stance controller
- q_{des} doing **inverse kinematics**, asking “for a given foot position (e.g. in a particular swing moment), **which joints bring my foot there?**”





Two lab sessions on **a complete model-based framework** for a legged quadruped:

- Friday: Test in simulation
 - State Estimation Tuning
 - Terrain Estimator
 - MPC Tuning
 - State Estimation Tuning

- Saturday: Test on a real robot
 - Periodic Gait Generator
 - Stance Control
 - Swing Control

You can find the exercise + explanation in here:

https://github.com/idra-lab/where_summer_school/blob/main/practical_sessions/practical_session_friday.pdf

- **30 minutes per exercise:** after each time slot, a solution/discussion for it will be shown on the slides
- **Please start from the first one!** See you in 30 min